

ECONOMETRICS I

Problem Set 2

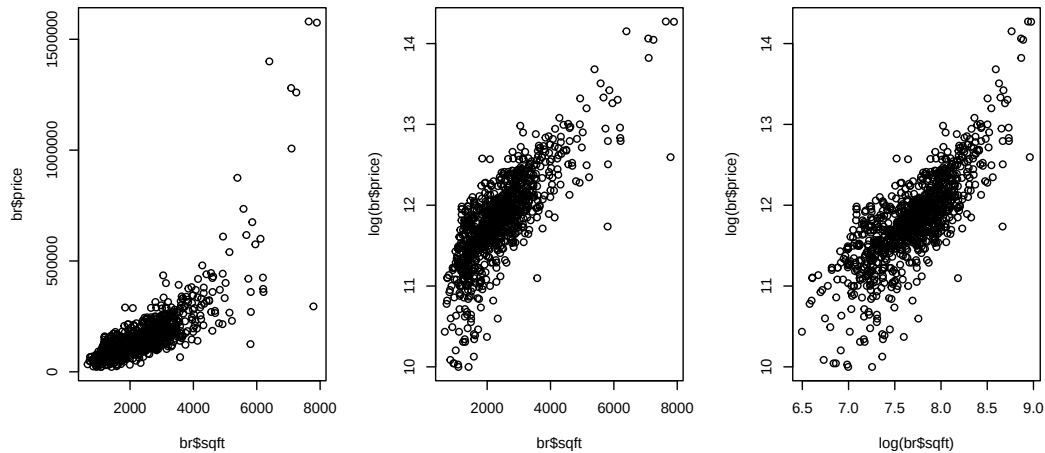
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November 3, 2025

1. **Application: real estate data.** Download `br.RData` from StudIP. This dataset (originally from Hill et al. 2018, *Principles of Econometrics*) contains data on 1080 houses sold in Baton Rouge, Louisiana, during mid-2005.

- (a) Plot `price` vs. `sqft`. What kind of model or transformation would the scatter suggest?

Solution:



The level-level plot is clearly non-linear. This calls for either a polynomial model or a transformation. The log-linear and log-log plots suggest that the latter transformation is more adequate.

- (b) Estimate the relationship of `price` vs. `sqft` with a level-level, a quadratic polynomial, a log-level, and a log-log model. You should get the results shown in table 1. What model offers the best fit? Does your graphical analysis lead to the same

conclusion?

Table 1: Result of question 1b

	<i>Dependent variable:</i>			
	price		log(price)	
	(1)	(2)	(3)	(4)
sqft	92.747*** (2.411)	-41.153*** (6.623)	0.0004*** (0.00001)	
I(sqft^2)		0.021*** (0.001)		
log(sqft)				1.006*** (0.025)
Constant	-60,861.460*** (6,110.187)	113,226.500*** (9,672.195)	10.839*** (0.025)	4.080*** (0.189)
Observations	1,080	1,080	1,080	1,080
R ²	0.579	0.703	0.625	0.607
<i>Note:</i>		*p<0.1; **p<0.05; ***p<0.01		

Solution: According to the R^2 , the best model is the quadratic one. However, visual inspection reveals that all but the log-log seem sensitive to outliers. Therefore, the recommended one would be the log-log specification.

- (c) You are considering buying a 2,000 square-foot property for \$ 120,000. Is that a good price? Answer according to each estimate of table 1.

Solution:

	(1)	(2)	(3)	(4)
x	2000	2000	2000	2000
$\hat{y}$	\$124,633	\$114,921	\$113,437	\$123,811
Gain/loss	\$4,633	-\$5,080	-\$6,563	\$3,811

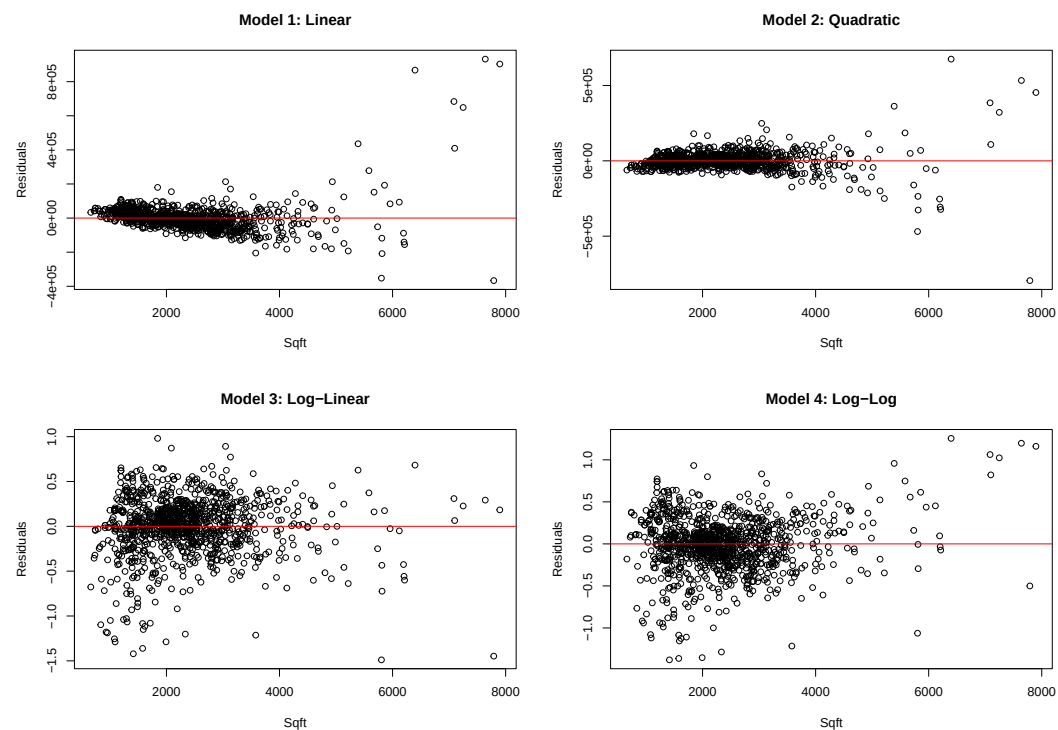
Models 2 and 3 suggest that the price is above average. Models 1 and 4 suggest the opposite.

- (d) Interpret the effect of size on price according to each estimate of of table 1.

Solution: (1) An extra square feet increase the price by \$ 92.75
 (2) An extra square feet increase the price by $-41.153 + 0.021 \cdot 2 \cdot sqft$ \$ USD.
 (3) An extra square feet increase the price by 0.04%.
 (4) A 1% increase in size increases the price by 1.006%

- (e) Do the regressions exhibit homoskedastic errors? Generate the residuals via, e.g., `br$resid1 <- residuals(model1)` and plot them against `sqft`. What is no longer valid in case of heteroskedasticity?

Solution:

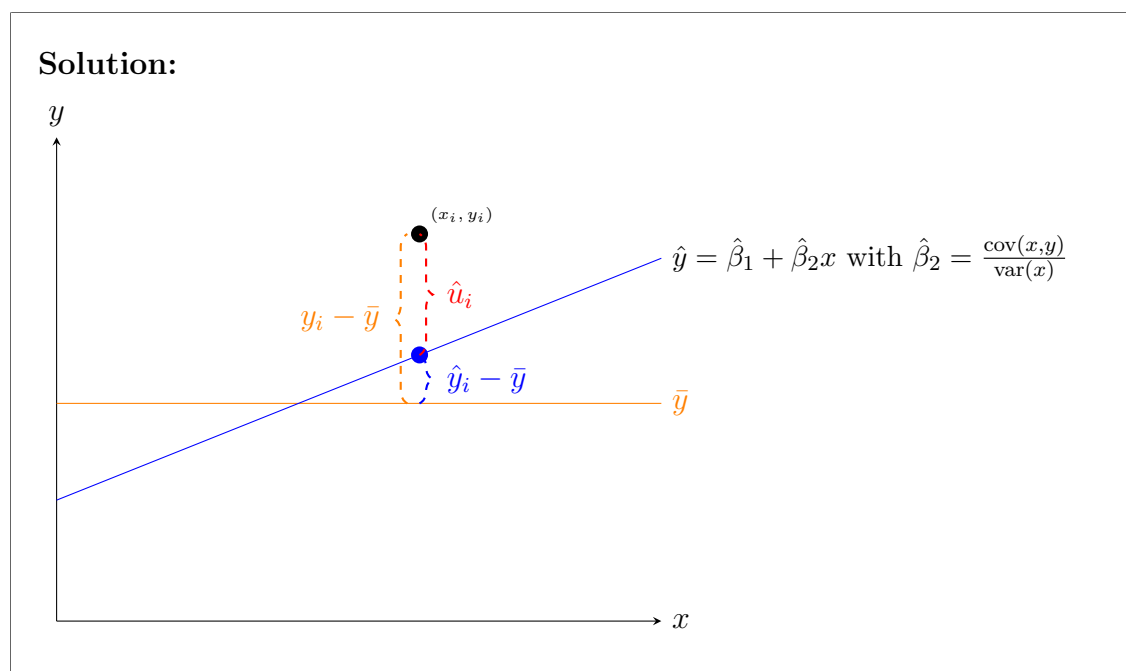


Models 1 and 2 are clearly heteroskedastic, 4 also to some extent. This means that the reported st. errors of the coefficients $\hat{s}(\hat{\beta}_j)$ are not properly computed and cannot be trusted. Only 3 appears to obey homoskedasticity.

2. R-squared

- (a) Explain graphically why the variance of y can be decomposed as the variance of $\hat{y}$

and that of $\hat{u}$.



(b) Show that

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{TSS} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{ESS} + \underbrace{\sum_{i=1}^n \hat{u}_i^2}_{SSR}$$

so that $0 \leq R^2 \leq 1$.

Solution:

Recall the two first order conditions $\frac{\partial SSR}{\partial \hat{\beta}_1} = 0$ and $\frac{\partial SSR}{\partial \hat{\beta}_2} = 0$ from problem set 1. They can be rewritten as

$$\sum_{i=1}^n \hat{u}_i = 0 \quad (1^*)$$

$$\sum_{i=1}^n x_i \hat{u}_i = 0 \quad (2^*)$$

We will use them below.

Let us expand TSS as follows:

$$\begin{aligned}
 \text{TSS} &= \sum_{i=1}^n (y_i - \bar{y})^2 \\
 &= \sum_{i=1}^n (\hat{y}_i - \bar{y} + \hat{u}_i)^2 \quad (\text{since } y_i = \hat{y}_i + \hat{u}_i) \\
 &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n \hat{u}_i^2 + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y}) \hat{u}_i
 \end{aligned}$$

Now examine the cross term:

$$\begin{aligned}
 \sum_{i=1}^n (\hat{y}_i - \bar{y}) \hat{u}_i &= \sum_{i=1}^n \hat{y}_i \hat{u}_i - \bar{y} \sum_{i=1}^n \hat{u}_i \\
 &= \sum_{i=1}^n (\hat{\beta}_1 + \hat{\beta}_2 x_i) \hat{u}_i - 0 \quad (\text{by (1*)}) \\
 &= \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_2 \sum_{i=1}^n x_i \hat{u}_i \\
 &= 0 + 0 = 0 \quad (\text{by (1*) and (2*)})
 \end{aligned}$$

Therefore:

$$\begin{aligned}
 \text{TSS} &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n \hat{u}_i^2 \\
 &= \text{ESS} + \text{RSS}
 \end{aligned}$$

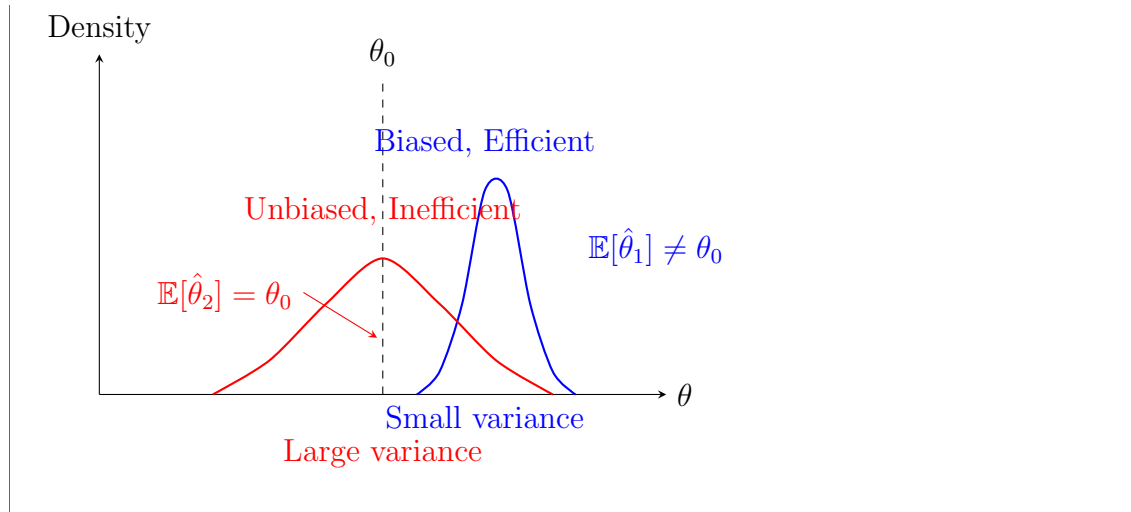
And R^2 is defined as:

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

3. **Unbiasedness.** We say that an estimator $\hat{\theta}$ of the population parameter θ is unbiased if, on average in repeating sampling, it coincides with its population counterpart—i.e., when $\mathbb{E}(\hat{\theta}) = \theta$.

- (a) Distinguish graphically between unbiasedness and efficiency.

Solution:



(b) Starting from the formula

$$\hat{\beta}_2 = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

of the simple (univariate) regression model, show that the condition $\mathbb{E}(u_i|x) = 0$ —or, alternatively, $\mathbb{E}[\text{cov}(u, x)] = 0$ —is required in addition to the assumption

$$y = \beta_1 + \beta_2 x + u$$

for $\mathbb{E}(\hat{\beta}_2|x) = \beta_2$ to hold.

Solution: By the rules described in the lecture's appendix:

$$\begin{aligned} \hat{\beta}_2 &= \frac{\text{cov}(x, x\beta_2)}{\text{var}(x)} + \frac{\text{cov}(x, u)}{\text{var}(x)} \\ &= \beta_2 \frac{\text{cov}(x, x)}{\text{var}(x)} + \frac{\text{cov}(x, u)}{\text{var}(x)} \\ &= \beta_2 \frac{\text{var}(x)}{\text{var}(x)} + \frac{\text{cov}(x, u)}{\text{var}(x)} \\ &= \beta_2 + \frac{\text{cov}(x, u)}{\text{var}(x)} = \beta_2 + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned}$$

Hence, we get $\mathbb{E}(\hat{\beta}_2) = \beta_2$ by assuming $\mathbb{E}[\text{cov}(u, x)] = 0$ or by assuming $\mathbb{E}(u_i|x) = 0$, so that $\mathbb{E}\left(\frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} | x\right) = \frac{\sum_{i=1}^n (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \mathbb{E}(u_i|x) = 0$.

(c) Starting from the formula

$$\hat{\beta} = (X'X)^{-1}X'y$$

of the multivariate regression model, show that the condition $\mathbb{E}(u|X) = 0$ —or, alternatively, $\mathbb{E}(X'u) = 0$ —is required in addition to the assumption

$$y = X\beta + u$$

for $\mathbb{E}(\hat{\beta}|X) = \beta$ to hold.

Solution:

$$\begin{aligned}
 \hat{\beta} &= (X'X)^{-1}X'y \\
 &= (X'X)^{-1}X'(X\beta + u) && \text{given } y = X\beta + u \\
 &= \beta + (X'X)^{-1}X'u \\
 \mathbb{E}(\hat{\beta}|X) &= \beta + \mathbb{E}[(X'X)^{-1}X'u|X] \\
 &= \beta + (X'X)^{-1}X'\mathbb{E}[u|X] \\
 &= \beta && \text{given } \mathbb{E}[u|X] = 0 \text{ or } \mathbb{E}[X'u] = 0
 \end{aligned}$$

4. **Estimator variance.** We define an estimator variance as $\mathbb{E}[(\hat{\theta} - \mathbb{E}(\hat{\theta}))^2]$ in repeating sample.

(a) Show that the variance of the slope $\hat{\beta}_2$ in a simple regression that we obtained in class,

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad (1)$$

can also be obtained from solving $\mathbb{E}(\hat{\beta}_2 - \mathbb{E}(\hat{\beta}_2))^2$.

Solution:

$$\begin{aligned}
 \text{var}(\hat{\beta}_2|x) &= \mathbb{E}[(\hat{\beta}_2 - \mathbb{E}[\hat{\beta}_2|x])^2|x] \\
 &= \mathbb{E}[(\hat{\beta}_2 - \beta_2)^2|x] && [\text{from (b) we know: } \mathbb{E}[\hat{\beta}_2|x] = \beta_2] \\
 &= \mathbb{E} \left[\left(\frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)^2 \middle| x \right] && [\text{also from question (b)}] \\
 &= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \mathbb{E} \left[\left(\sum_{i=1}^n (x_i - \bar{x})u_i \right)^2 \middle| x \right]
 \end{aligned}$$

Now expand the square:

$$\begin{aligned}
 \mathbb{E} \left[\left(\sum_{i=1}^n (x_i - \bar{x})u_i \right)^2 \middle| x \right] &= \mathbb{E} \left[\sum_{i=1}^n \sum_{j=1}^n (x_i - \bar{x})(x_j - \bar{x})u_i u_j \middle| x \right] \\
 &= \sum_{i=1}^n \sum_{j=1}^n (x_i - \bar{x})(x_j - \bar{x}) \mathbb{E}[u_i u_j | x]
 \end{aligned}$$

Note that, since $\mathbb{E}[u_i|x] = 0$ for all i , the following holds:

$$\text{cov}(u_i, u_j|x) = \mathbb{E}[(u_i - \mathbb{E}[u_i|x])(u_j - \mathbb{E}[u_j|x])|x] = \mathbb{E}[u_i u_j|x]$$

Therefore:

- $\mathbb{E}[u_i u_j|x] = \text{cov}(u_i, u_j|x) = 0$ for $i \neq j$ (no autocorrelation)
- $\mathbb{E}[u_i^2|x] = \text{var}(u_i|x) = \sigma^2$ (homoskedasticity)

Using these expressions, the expanded squared term has expectation

$$\begin{aligned} \mathbb{E} \left[\left(\sum_{i=1}^n (x_i - \bar{x}) u_i \right)^2 \middle| x \right] &= \sum_{i=1}^n (x_i - \bar{x})^2 \mathbb{E}[u_i^2|x] \\ &= \sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2 \end{aligned}$$

Therefore, going back to our variance:

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

(b) Show that, for a random vector $a_{(n \times 1)}$, it holds that

$$V[a] \equiv \mathbb{E} [(a - \mathbb{E}[a])(a - \mathbb{E}[a])'] = \begin{bmatrix} \text{var}[a_1] & \text{cov}[a_1, a_2] & \cdots & \text{cov}[a_1, a_n] \\ \text{cov}[a_2, a_1] & \text{var}[a_2] & \cdots & \text{cov}[a_2, a_n] \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}[a_n, a_1] & \text{cov}[a_n, a_2] & \cdots & \text{var}[a_n] \end{bmatrix}$$

Note: we call $V[a]$ the **variance-covariance matrix** of a .

Solution:

$$\begin{aligned}
V[a] &= \mathbb{E}[(a - \mathbb{E}[a])(a - \mathbb{E}[a])'] \\
&\quad n \times n \\
&= \mathbb{E} \left[\begin{bmatrix} (a_1 - \mathbb{E}[a_1]) \\ \vdots \\ (a_n - \mathbb{E}[a_n]) \end{bmatrix} \begin{bmatrix} (a_1 - \mathbb{E}[a_1]) & \dots & (a_n - \mathbb{E}[a_n]) \end{bmatrix} \right] \\
&= \begin{bmatrix} \mathbb{E}[(a_1 - \mathbb{E}[a_1])(a_1 - \mathbb{E}[a_1])] & \dots & \mathbb{E}[(a_1 - \mathbb{E}[a_1])(a_n - \mathbb{E}[a_n])] \\ \mathbb{E}[(a_2 - \mathbb{E}[a_2])(a_1 - \mathbb{E}[a_1])] & \dots & \mathbb{E}[(a_2 - \mathbb{E}[a_2])(a_n - \mathbb{E}[a_n])] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[(a_n - \mathbb{E}[a_n])(a_1 - \mathbb{E}[a_1])] & \dots & \mathbb{E}[(a_n - \mathbb{E}[a_n])(a_n - \mathbb{E}[a_n])] \end{bmatrix} \\
&= \begin{bmatrix} \text{var}[a_1] & \text{cov}[a_1, a_2] & \dots & \text{cov}[a_1, a_n] \\ \text{cov}[a_2, a_1] & \text{var}[a_2] & \dots & \text{cov}[a_2, a_n] \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}[a_n, a_1] & \text{cov}[a_n, a_2] & \dots & \text{var}[a_n] \end{bmatrix}
\end{aligned}$$

- (c) Show that the assumptions of homoskedasticity and no-autocorrelation of the error can be summarized as

$$V[u|X] = \mathbb{E}[uu'|X] = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix} = \sigma^2 I_{n \times n} \quad (2)$$

Solution:

$$\begin{aligned}
V[u|X] &= \mathbb{E}[(u - \mathbb{E}[u])(u - \mathbb{E}[u])'|X] = \mathbb{E}[uu'|X] = \mathbb{E} \left[\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} u_1 & \dots & u_n \end{bmatrix} | X \right] \\
&= \begin{bmatrix} E[u_1^2|X] & E[u_1 u_2|X] & \dots & E[u_1 u_n|X] \\ E[u_2 u_1|X] & E[u_2^2|X] & \dots & E[u_2 u_n|X] \\ \vdots & \vdots & \ddots & \vdots \\ E[u_n u_1|X] & E[u_n u_2|X] & \dots & E[u_n^2|X] \end{bmatrix}
\end{aligned}$$

- (d) Show that the matrix analog of (1) is

$$V(\hat{\beta}|X) = \begin{bmatrix} \text{var}(\hat{\beta}_1|X) & & \\ & \ddots & \\ & & \text{var}(\hat{\beta}_k|X) \end{bmatrix} = \sigma^2 (X'X)^{-1}$$

Solution: Recall that under $\mathbb{E}(X'u) = 0$ it holds that

$$\mathbb{E}(\hat{\beta}) = \beta + (X'X)^{-1}X'u \quad (3)$$

Hence.

$$\begin{aligned} V[\hat{\beta}|X] &= \mathbb{E} \left[(\hat{\beta} - \mathbb{E}[\hat{\beta}|X])(\hat{\beta} - \mathbb{E}[\hat{\beta}|X])' | X \right] \\ &= \mathbb{E} \left[((X'X)^{-1}X'u)((X'X)^{-1}X'u)' | X \right] && \text{using (3)} \\ &= \mathbb{E} \left[((X'X)^{-1}X'u)(u'X(X'X)^{-1}) | X \right] \\ &= \mathbb{E} \left[(X'X)^{-1}X'uu'X(X'X)^{-1} | X \right] \\ &= (X'X)^{-1}X'\mathbb{E}[uu'|X]X(X'X)^{-1} \\ &= (X'X)^{-1}X'\sigma^2 I_{n \times n}X(X'X)^{-1} && \text{given (2)} \\ &= \sigma^2(X'X)^{-1} \end{aligned}$$

- (e) Make an R script computing $\hat{V}[\hat{\beta}|X] = \hat{\sigma}^2(X'X)^{-1}$, where $\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{n-k} = \frac{\hat{u}'\hat{u}}{n-k}$.

Solution: See ‘script PS2 OLS-Matrices’ uploaded to folder ‘Lecture 5’